CO_2 EMISSIONS FROM VEHICLES IN CANADA

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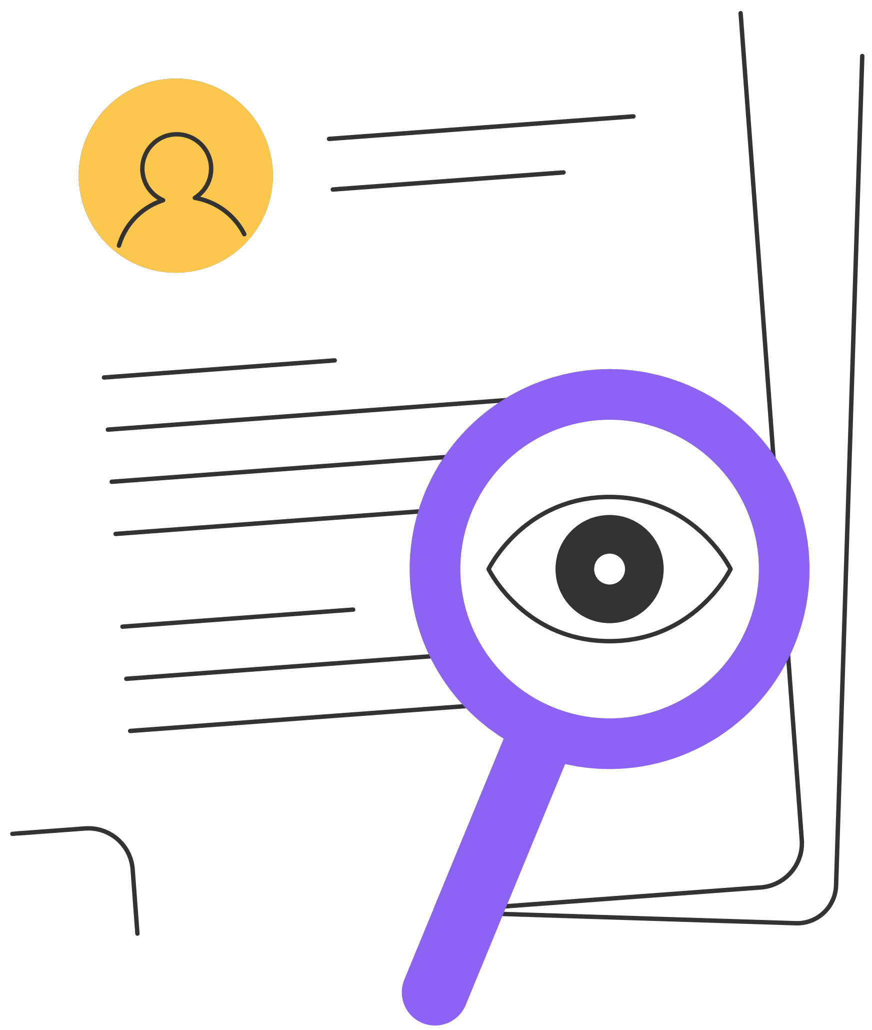
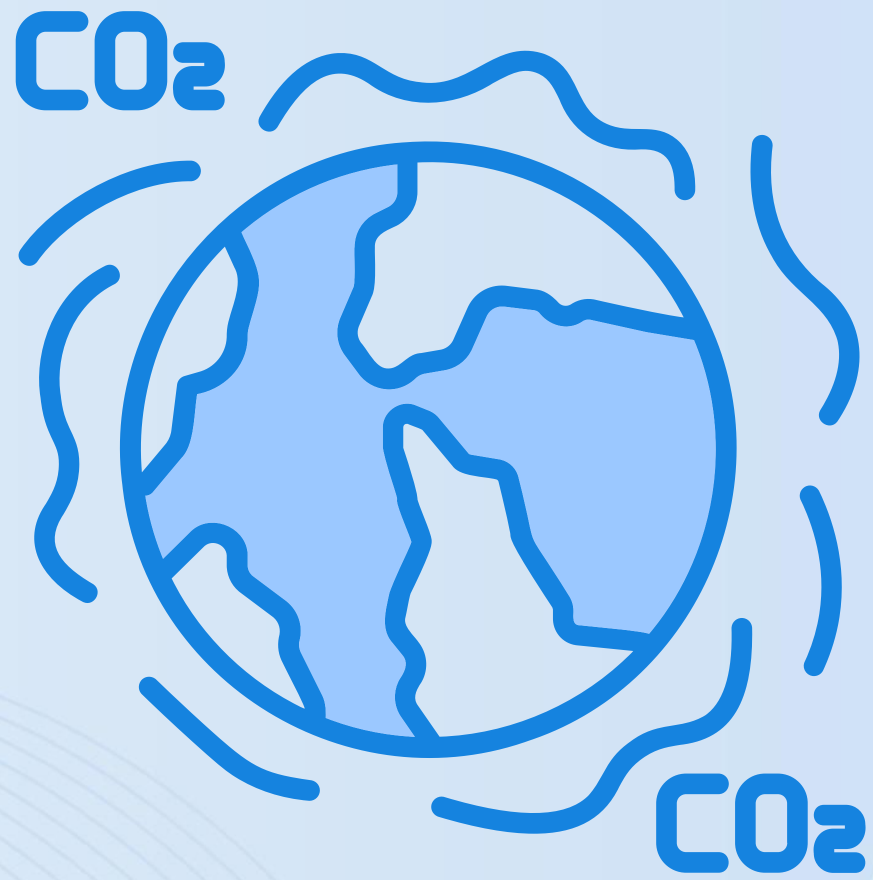
Group 07 | ST 3008 Applied Statistical Models

Introduction & Literature Review



Background

- CO_2 is essential but harmful in excess
- Human activity → Rising CO_2 levels
- Transportation is a major contributor
- Consequences: Global warming, rising seas, disasters
- Push towards sustainable transport (EVs, clean fuels)



Dataset

- Source: Canadian Government's fuel consumption rating dataset
- 7,385 records, 2,053 car models, 42 brands
- Features: engine size, cylinders, fuel type, transmission, fuel consumption, CO_2 emissions

Objectives

- Identify key factors affecting CO_2 emissions
- Build multiple linear regression model
- Build binary logistic regression model (high vs. low emissions)

Literature Review – Key Studies

- Gurcan (2024): 18 algorithms → Ensemble models best (R^2 ≈ 99%)
- Abdullah & Theyazn (2023): BiLSTM outperformed LSTM (R^2 = 93.78)
- Mohammad (2025): CarbonMLP model (R^2 = 99.38) + SHAP analysis

Common Findings

Strongest predictors

- Engine size
- Fuel consumption (city, highway, combined)
- Cylinder count

Consistent results across regression, ensemble, and deep learning methods

Descriptive Analysis of CO_2 Emissions

Data and Variables

- The data is from the Canadian government's fuel consumption ratings program, spanning 7 years (2014-2020) with 7,385 records.

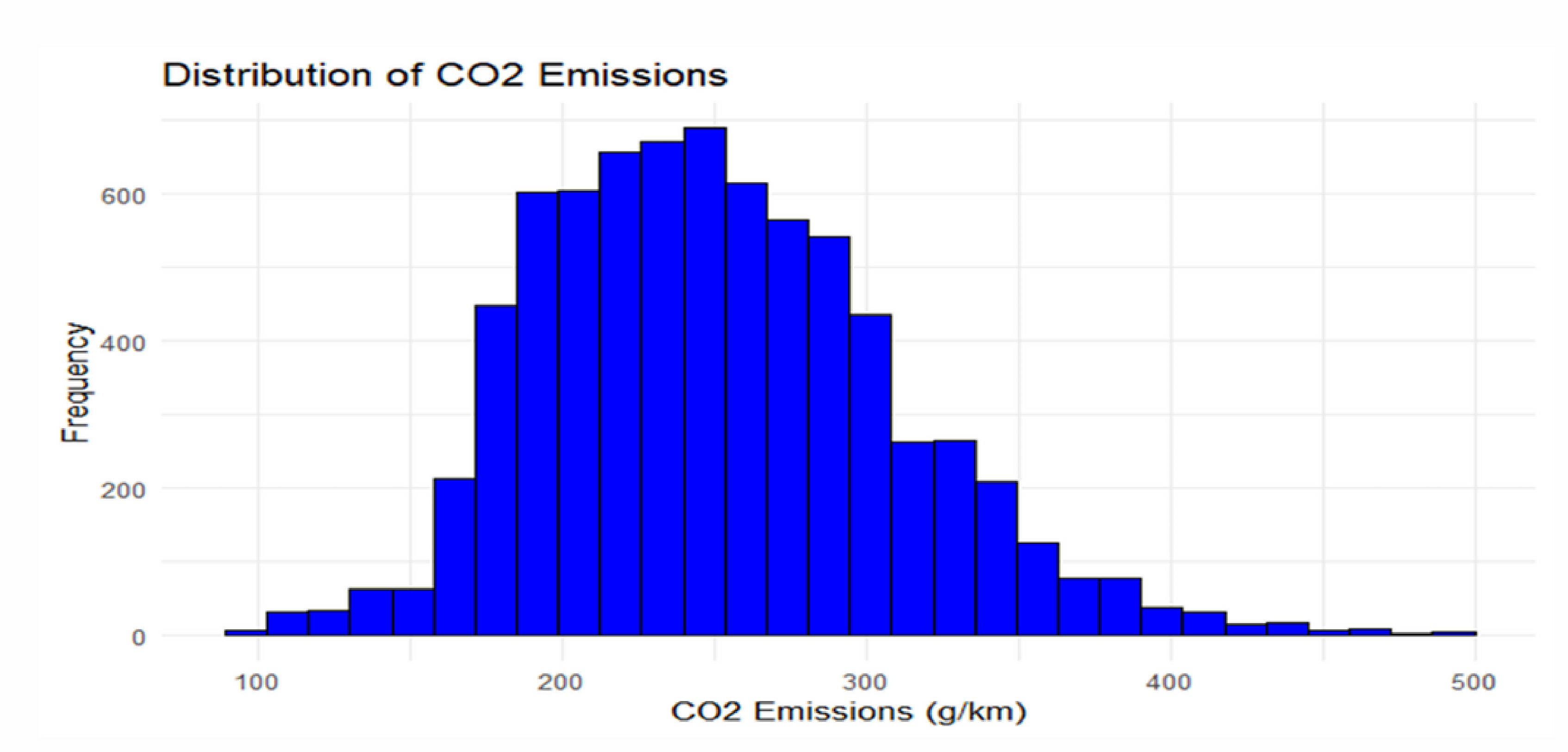
- We considered variables such as Engine Size (L), Cylinders, Fuel Type, Transmission, Vehicle Class, Combined Fuel Consumption (L/100km), and CO_2 Emissions (g/km).

- To make our models simpler, we grouped some categorical variables.
  - For example, we grouped the original transmission types into four: Automatic, Manual, Variable and Automated Manual.

- We discarded some variables that were not essential or were redundant.
  - For example we used Fuel Consumption Comb to measure fuel use and did not use the separate Fuel Consumption City and Fuel Consumption Hwy variables since they were used to calculate the combined value.

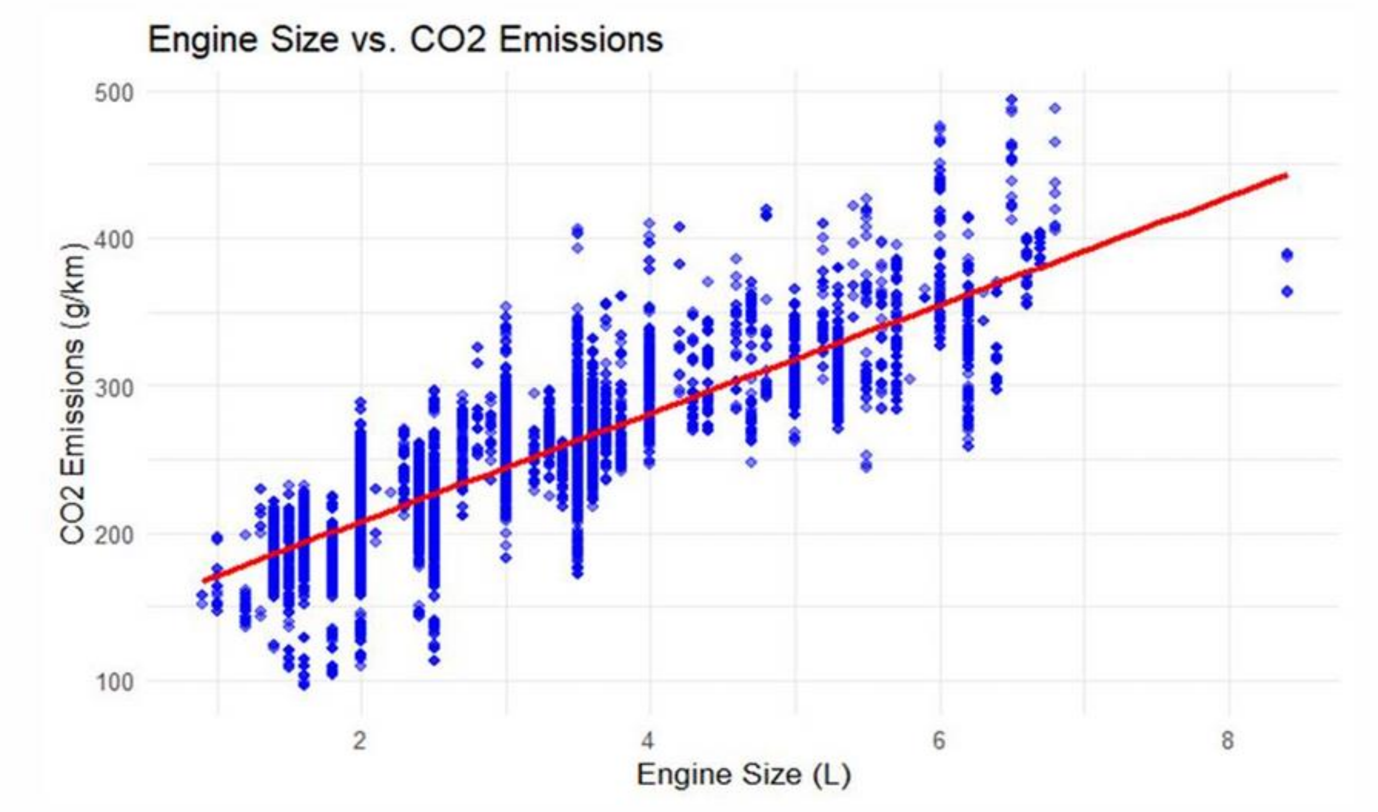
Exploratory Data Analysis

- Distribution of CO_2 Emissions



The distribution is right-skewed, with most vehicles emitting between 200 and 300 g/km.

- Engine Size vs. CO_2 Emissions

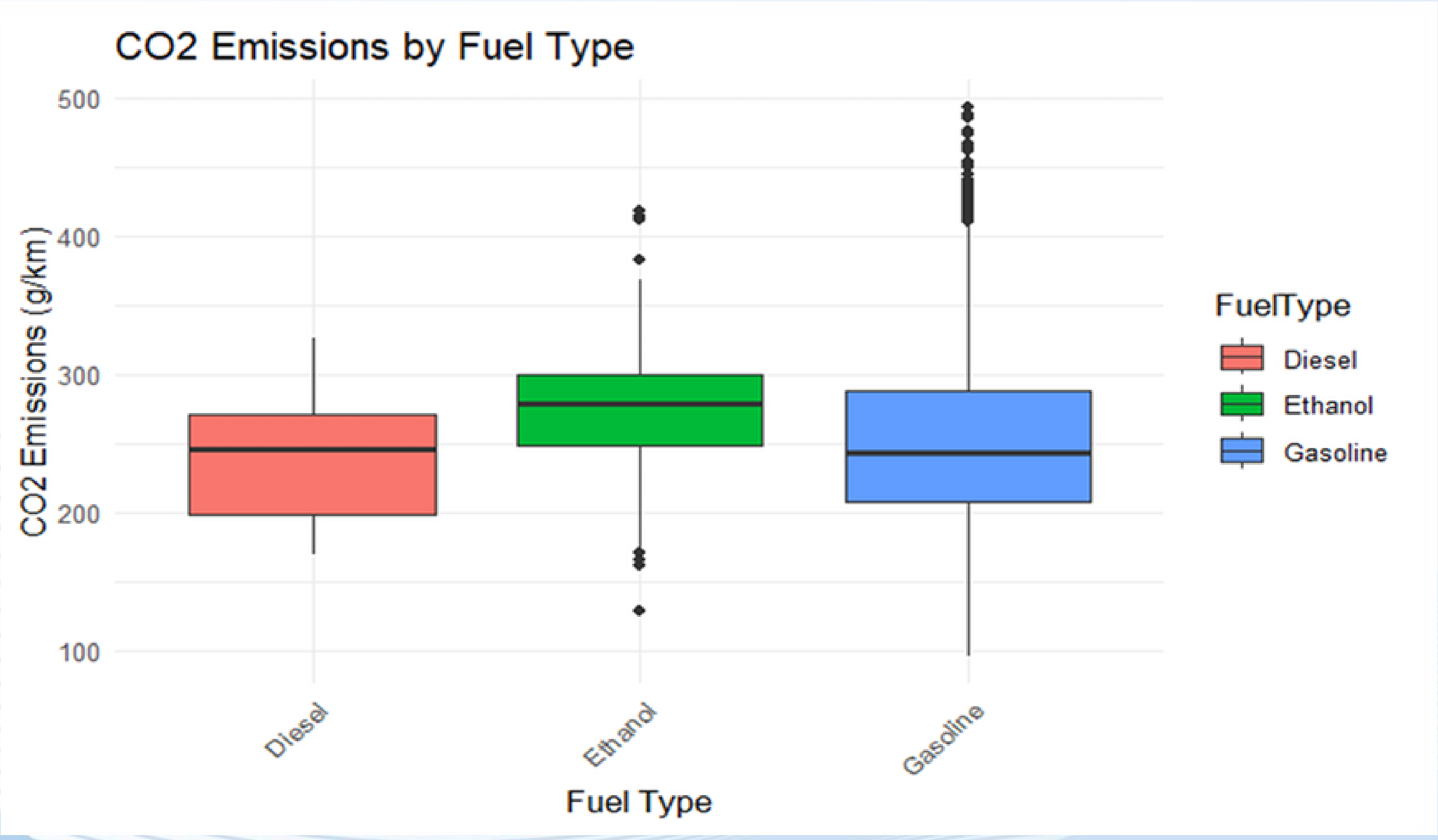


There is a strong positive linear relationship: as engine size increases, so do CO_2 emissions

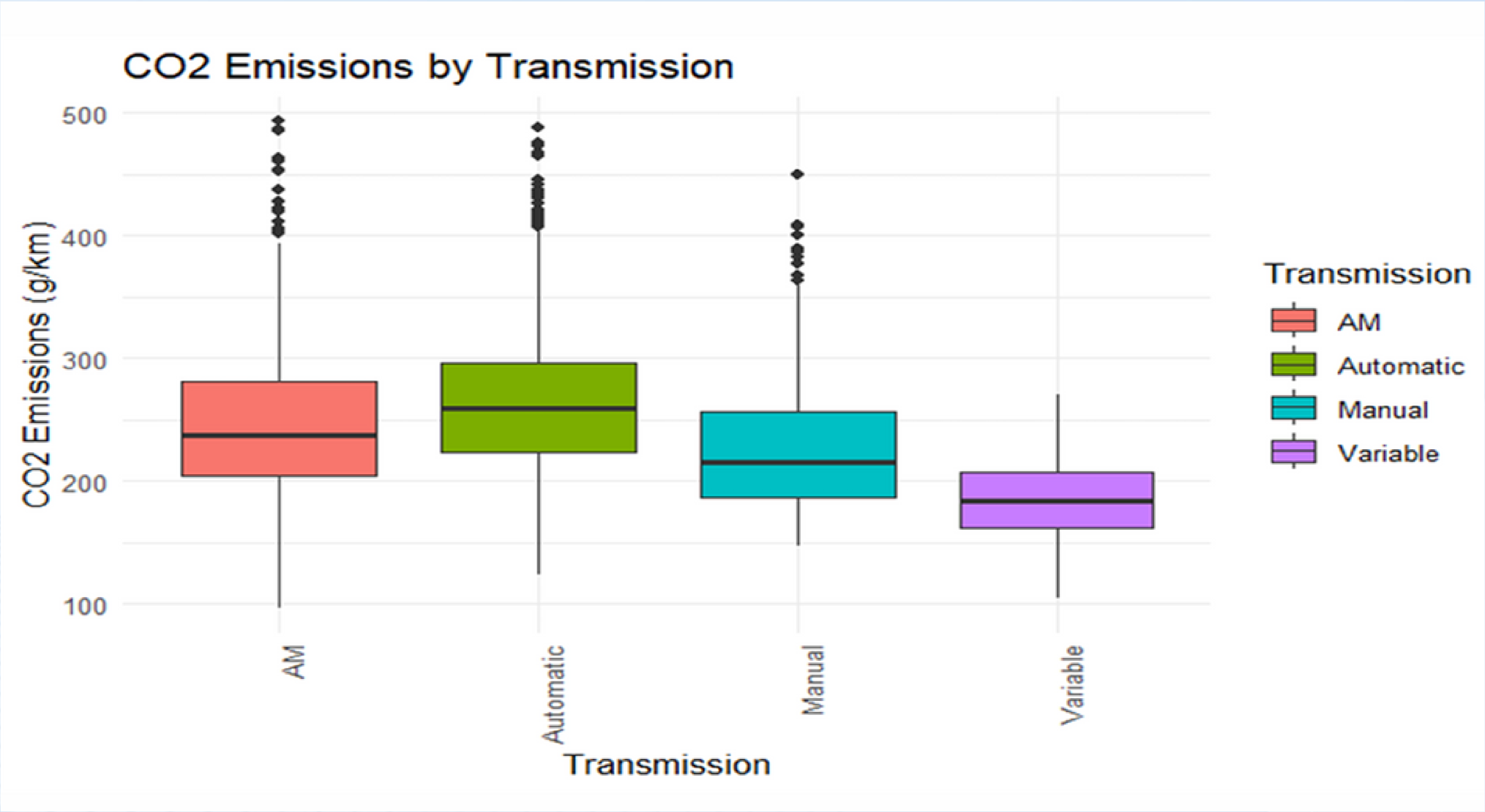
Exploratory Data Analysis

Emissions by Fuel Type & Transmission

- By Fuel Type



- By Transmission



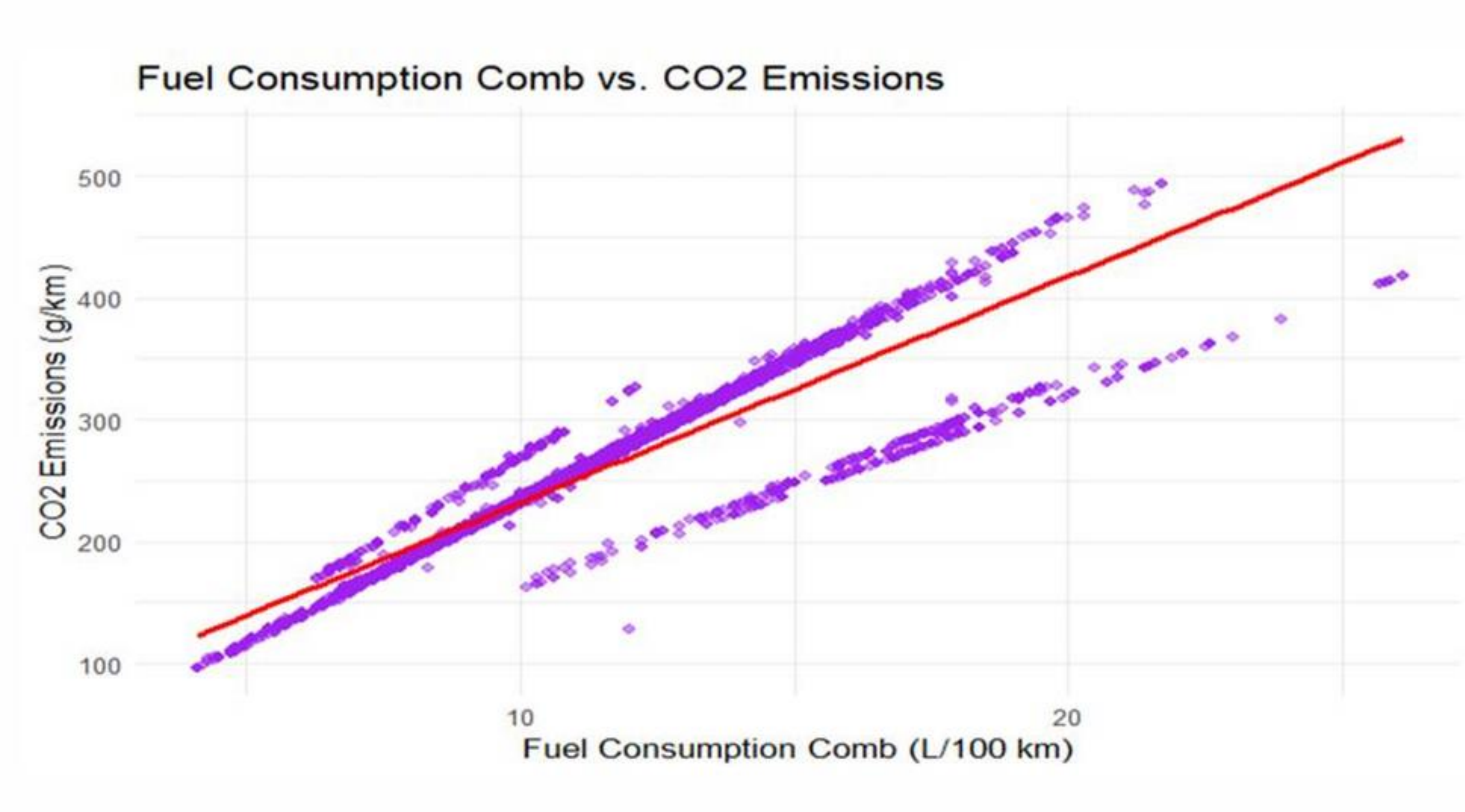
Ethanol-powered vehicles have the highest median emissions, while the gasoline category shows the widest spread.

Automatic transmissions have the highest median CO_2 emissions, and variable transmissions have the lowest.

Exploratory Data Analysis

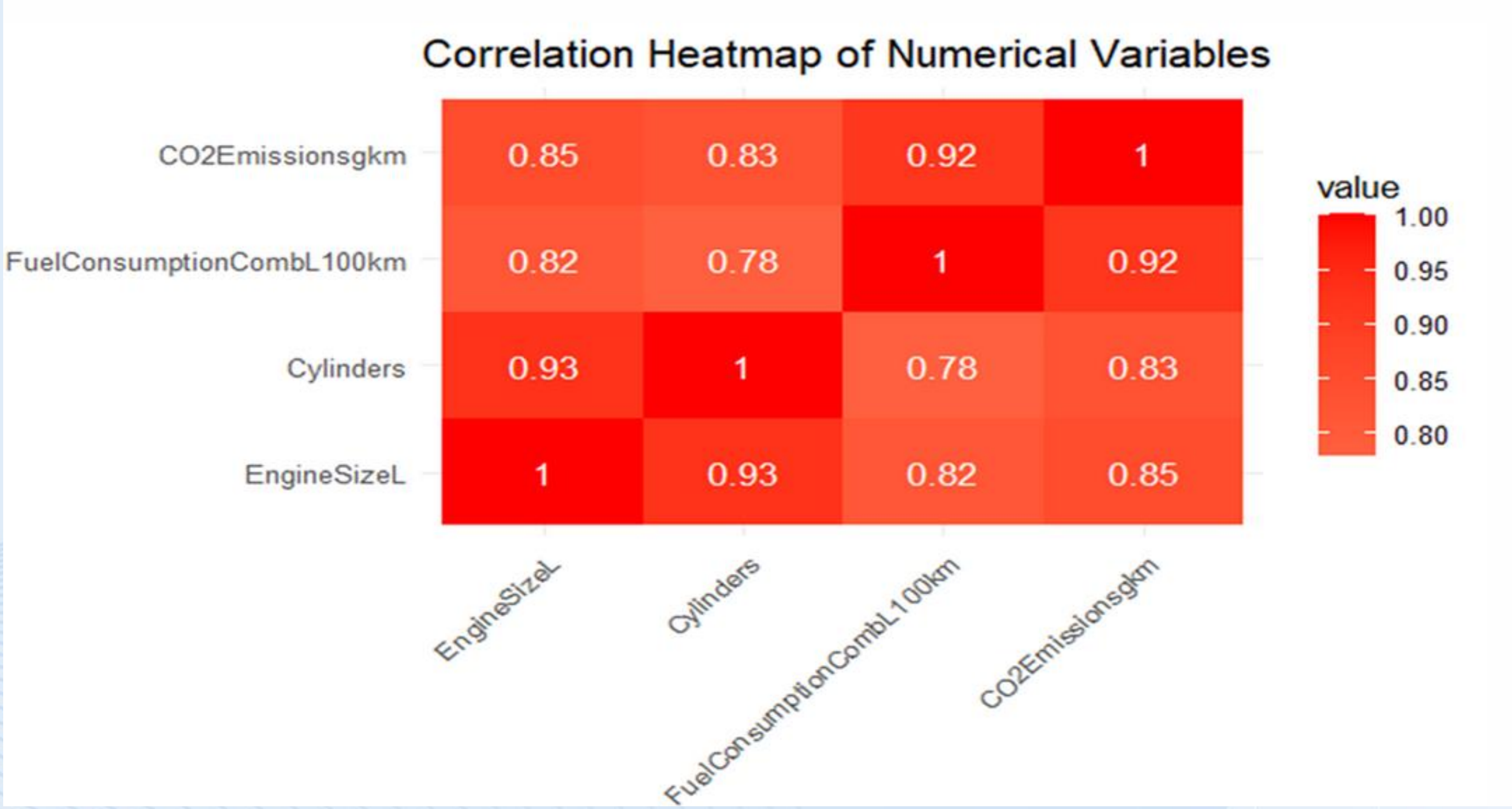
Uncovering the Strongest Links

- Fuel Consumption vs. Emissions



A very strong positive linear relationship exists, visually confirming that higher fuel consumption leads to higher CO_2 emissions.

- Correlation of Numerical Variables



The heatmap shows a very strong correlation between CO_2 Emissions and Fuel Consumption Comb (0.92).

Summary of Findings

- The descriptive analysis showed a strong positive relationship between CO_2 emissions and factors like Fuel Consumption, Engine Size and Cylinders.

- We also found that categorical variables such Fuel Type and Transmission significantly influence the CO_2 emissions.

- These insights provide a solid foundation for the next stage of our project which is building and analyzing the regression models.

Multiple Linear Regression Model for CO_2 Emissions Prediction

Model Overview

- Purpose: Predict CO_2 emissions (g/km) using vehicle characteristics

- Variables:
  - Categorical: Vehicle Class, Transmission, Fuel Type
  - Quantitative: Engine Size (L), Cylinders, Combined Fuel Consumption (L/100km)

- Goal: Identify significant predictors of CO_2 emissions

Dummy Variables

- Vehicle Class
  - Large (Reference)
  - Medium → $VClass_M$
  - Small → $VClass_S$
- Fuel Type
  - Gasoline (Reference)
  - Deisel → $Fuel_D$
  - Ethanol → $Fuel_E$
- Transmission
  - Manual (Reference)
  - Automatic → $Trans_A$
  - Variable → $Trans_V$
  - Automated Manual → $Trans_{AM}$

Model Building

1. Started with the Null Model
2. Iteratively added significant variables using P - value
3. At each step evaluated model improvement using F-value
4. Repeated the process until no further variables significantly improve the model

Model Evaluation at Each Iterations

H_0 : reduced model is adequate
H_1 : full model is adequate

Test Statistic

$$F = \frac{[ESS(RM) - ESS(FM)]/(p-k)}{ESS(FM)/(n-p-1)}$$

- Tested at 5% level significance

Fitted Model by the Forward Selection

$$
\begin{aligned}
\widehat{\text{CO2Emissionsgkm}} = {} & 8.57518 + 22.01761 \cdot \text{FuelConsumptionCombL100km} + 29.20363 \cdot \text{Fuel}_D - 111.77227 \cdot \text{Fuel}_E \\
& + 1.03097 \cdot \text{Cylinders} + 0.57029 \cdot \text{Trans}_{AM} + 0.54480 \cdot \text{Trans}_A \\
& - 2.19289 \cdot \text{Trans}_V - 0.97910 \cdot \text{VClass}_M - 1.46768 \cdot \text{VClass}_S
\end{aligned}
$$

```
Residual standard error: 5.376 on 7371 degrees of freedom
Multiple R-squared:  0.9915,    Adjusted R-squared:  0.9915
F-statistic: 9.55e+04 on 9 and 7371 DF,  p-value: < 2.2e-16
```

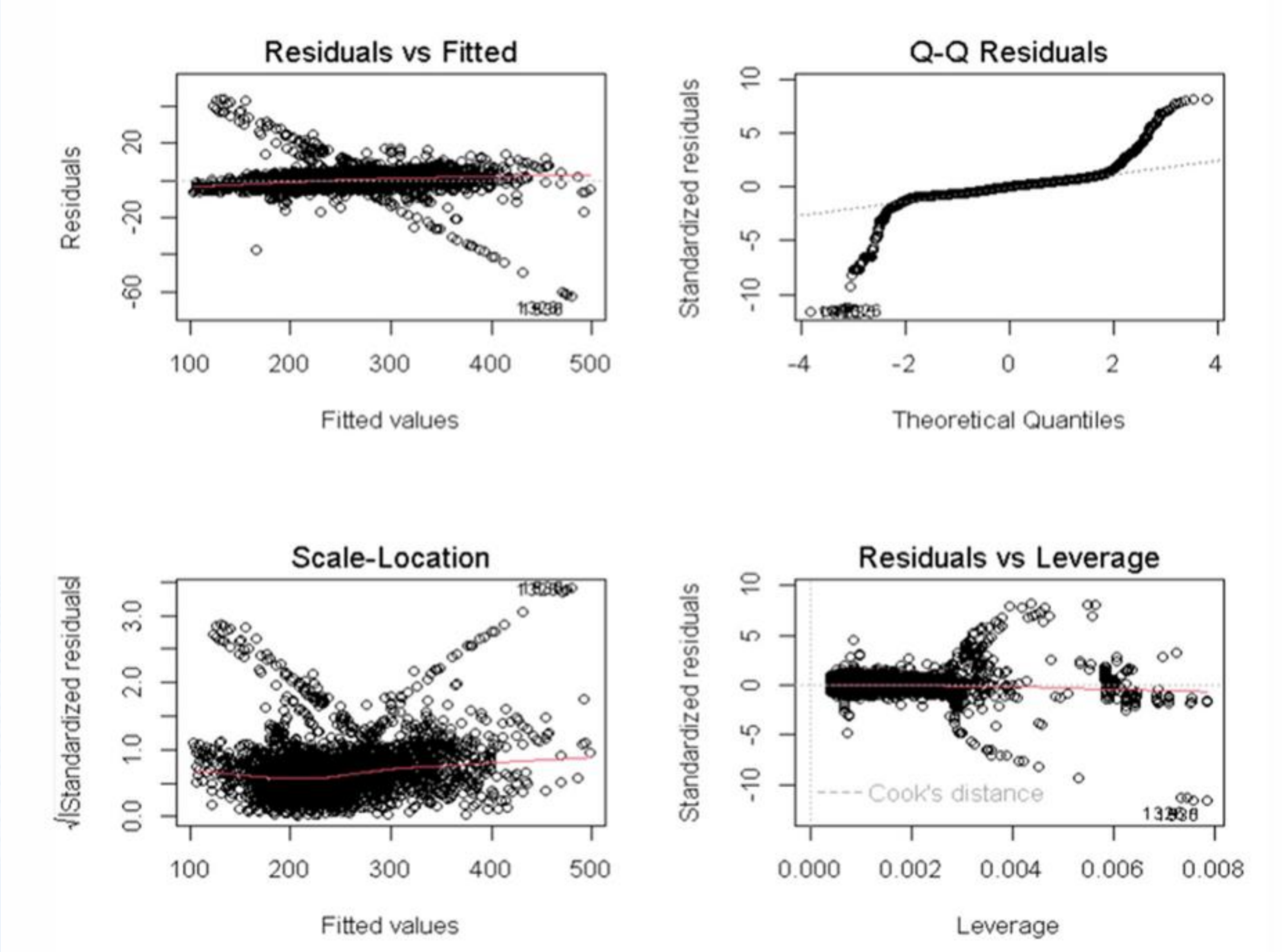
Model Adequacy Checking

Residuals vs Fitted Plot: Residuals scatter randomly around zero, indicating a good model fit, with a slight upward trend and minor deviations from linearity.

Q-Q Residuals Plot: Residuals mostly follow the reference line, showing near-normal distribution, but S-shaped deviations at extremes suggest slight non-normality.

Scale-Location Plot: Residuals are randomly scattered, supporting homoscedasticity, with minimal U-shaped curvature and negligible trends.

Residuals vs Leverage Plot: Residuals are mostly random, but a shallow U-shaped curve at higher leverage indicates some influential points.



Model Adequacy Checking

Model Adjustments

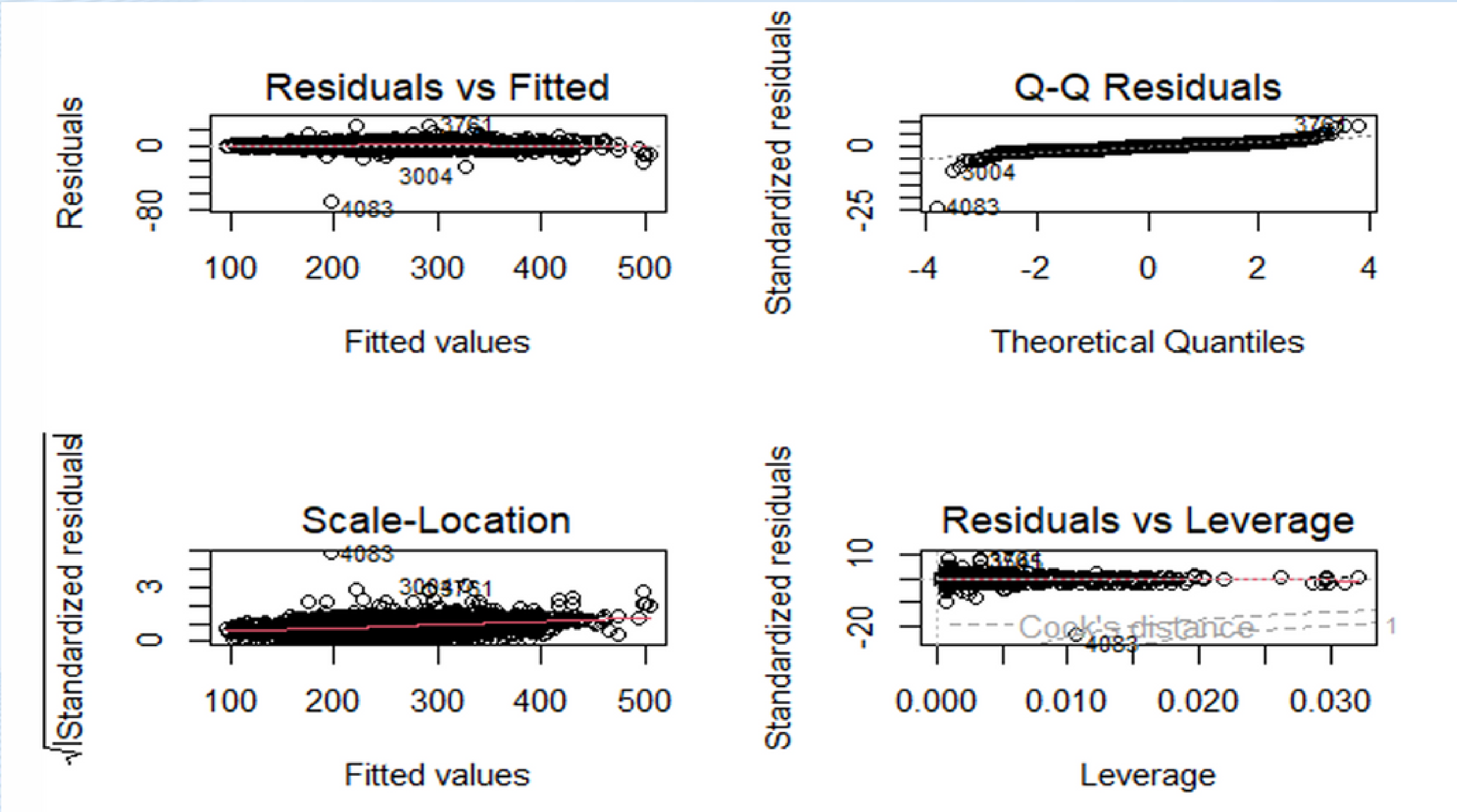
Log transformation, variable centering, and polynomials were tried but failed to improve the model. Then tried adding interactions and FuelConsumptionCombL100km*Fuel Type improved the model.

Residuals vs Fitted Plot: No curved patterns, suggesting linearity is satisfied.

Q-Q Plot: Residuals are normally distributed, meeting normality assumption

Scale-Location Plot: Approximately flat red line, indicating constant variance (homoscedasticity).

- There is minimum multicollinearity because all standard GIVF values are less than 10.



	GVIF	Df	GVIF^(1/(2*Df))
FuelConsumptionCombL100km	6.230441	1	2.496085
FuelType	1148.263118	2	5.821171
Transmission	1.563514	3	1.077334
Cylinders	3.966145	1	1.991518
VehicleClass	1.659734	2	1.135036
FuelConsumptionCombL100km:FuelType	1221.653434	2	5.912035

Final Adequate Model

$$
\begin{aligned}
\widehat{\text{CO2Emissionsgkm}} = {} & 1.98802 + 22.97367 \cdot \text{FuelConsumptionCombL100km} - 1.18938 \cdot \text{Fuel}_D + 3.41447 \cdot \text{Fuel}_E \\
& + 0.33176 \cdot \text{Trans}_{AM} + 0.18537 \cdot \text{Trans}_A - 0.48247 \cdot \text{Trans}_V + 0.33498 \cdot \text{Cylinders} \\
& - 0.48759 \cdot \text{VClass}_M - 0.88219 \cdot \text{VClass}_S \\
& + 3.63838 \cdot (\text{FuelConsumptionCombL100km} \times \text{Fuel}_D) \\
& - 7.11238 \cdot (\text{FuelConsumptionCombL100km} \times \text{Fuel}_E)
\end{aligned}
$$

```
Residual standard error: 2.926 on 7369 degrees of freedom
Multiple R-squared:  0.9975,    Adjusted R-squared:  0.9975
F-statistic: 2.655e+05 on 11 and 7369 DF,  p-value: < 2.2e-16
```

Estimates Interpretation

Intercept ⟶ 1.99 g/km (baseline emssion)

FuelConsumptionCombL100km Unit increase ⟶ 22.97 g/km ↑

Cylinders Unit increase ⟶ 0.33 g/km ↑

Deisel ⟶ 1.19 g/km ↓

Ethanol ⟶ 3.41 g/km ↑

Automated Manual ⟶ 0.33 g/km ↑

Auto ⟶ 0.19 g/km ↑

Variable ⟶ 0.48 g/km ↓

Medium ⟶ 0.49 g/km ↓

Small ⟶ 0.88 g/km ↓

FuelConsumptionCombL100km Unit increase and Deisel ⟶ 3.64 g/km ↑

FuelConsumptionCombL100km Unit increase and Ethanol ⟶ 7.11 g/km ↓

Logistic Regression Model for CO_2 Emissions Prediction

Model Overview

- Objective: To predict whether a vehicle has high or low CO_2 emissions based on its characteristics.

- Variables:
  - Categorical: Vehicle Class, Transmission, Fuel Type
  - Quantitative: Engine Size (L), Cylinders, Combined Fuel Consumption (L/100km)
  - Response variable: Categorical(High,Low)

Data Preparation

In this model, CO2Emissionsgkm was converted into a binary variable based on its median value.

High (coded as 1)	**CO2Emissionsgkm => median (CO2Emissionsgkm)**
Low (coded as 0)	**CO2Emissionsgkm < median (CO2Emissionsgkm)**

Why use median value for creating binary variable?

The median divides the data set so that 50% of values are below and 50% are above. This ensures that both categories have enough observations for analysis and avoid imbalanced groups

```
summary(CO2Emissionsgkm)
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 96.0   208.0   246.0   250.5   288.0   493.0
```

Model Building

Forward selection method was used to build the model

1. Start with null model (intercept only)

2. At each step, test candidate variables for statistically significant improvements in model fit using reductions in deviance (reduction in AIC value)

3. The predictor that provides the largest and statistically significant improvement (minimum AIC, minimum deviance of residuals) was included in the model

4. Repeat the process until no further variables significantly improve the model (no reduced deviance or AIC)

Model comparison at each iteration

H_0 : Reduced model is adequate
H_1 : Full model is adequate

Test statistic

ΔD = "Difference in residual deviances Between the reduced and full models."

Critical value

$C.V = X^2_{(5\%,\, df)}$

df = (number of parameters in full model) - (number of parameters in reduced model)

Then, if $\Delta D > C.V$ reject H_0

That means , the newly added variable is statistically significant.

Model in each iteration

1	**Null**	• ***Logit(p) ~ 1*** • Residual deviance 10230 on 7380 degrees of freedom
2	**FuelConsumption CombL100km**	• ***Logit(p) ~ FuelConsumptionCombL100km*** • Residual deviance 2861.7 on 7379 degrees of freedom
3	**Fuel Type**	• ***Logit(p) ~ FuelConsumptionCombL100km+Fuel type*** • Residual deviance 474.38 on 7377 degrees of freedom

4	**Vehicle Class**	• ***Logit(p) ~ FuelConsumptionCombL100km+Fuel type+ vehicle class*** • Residual deviance 448.67 on 7375 degrees of freedom
5	**Transmission**	• ***Logit(p) ~ FuelConsumptionCombL100km+Fuel type+ vehicle class + Transmission*** • Residual deviance 438.42 on 7372 degrees of freedom

During forward selection, Cylinders and Engine Size were tested but showed no significant reduction in deviance.
Therefore, Cylinders and Engine Size were excluded from the final model.

$$Logit\,(p) = ln\left(\frac{p}{1-p}\right)$$

P = Probability(CO2Emissionsgkm = high)

Fitted Model

$$\ln\left(\frac{\hat{P}}{1-\hat{P}}\right) = -175.434216.6041 \cdot \mathrm{FuelConsumptionCombL100km}$$
$$+ 21.3125 \cdot \mathrm{Fuel}_D - 70.7008 \cdot \mathrm{Fuel}_E$$
$$- 1.2783 \cdot \mathrm{VClass}_M - 1.5818 \cdot \mathrm{VClass}_S$$
$$+ 1.4475 \cdot \mathrm{Trans}_{AM} + 0.3211 \cdot \mathrm{Trans}_A - 0.5468 \cdot \mathrm{Trans}_V$$

$\hat{P}$ is the estimated proportion for high co2Emissionsgkm

Goodness of fit test

H_0 = Model fits the data well
H_1 = Model does not fit the data well

Test statistic(deviance)=438.42
degrees of freedom=7372
critical value at 5% significance level= 7572.857

At 5% significance level,
test statistic(deviance) < critical value, do not reject H_0

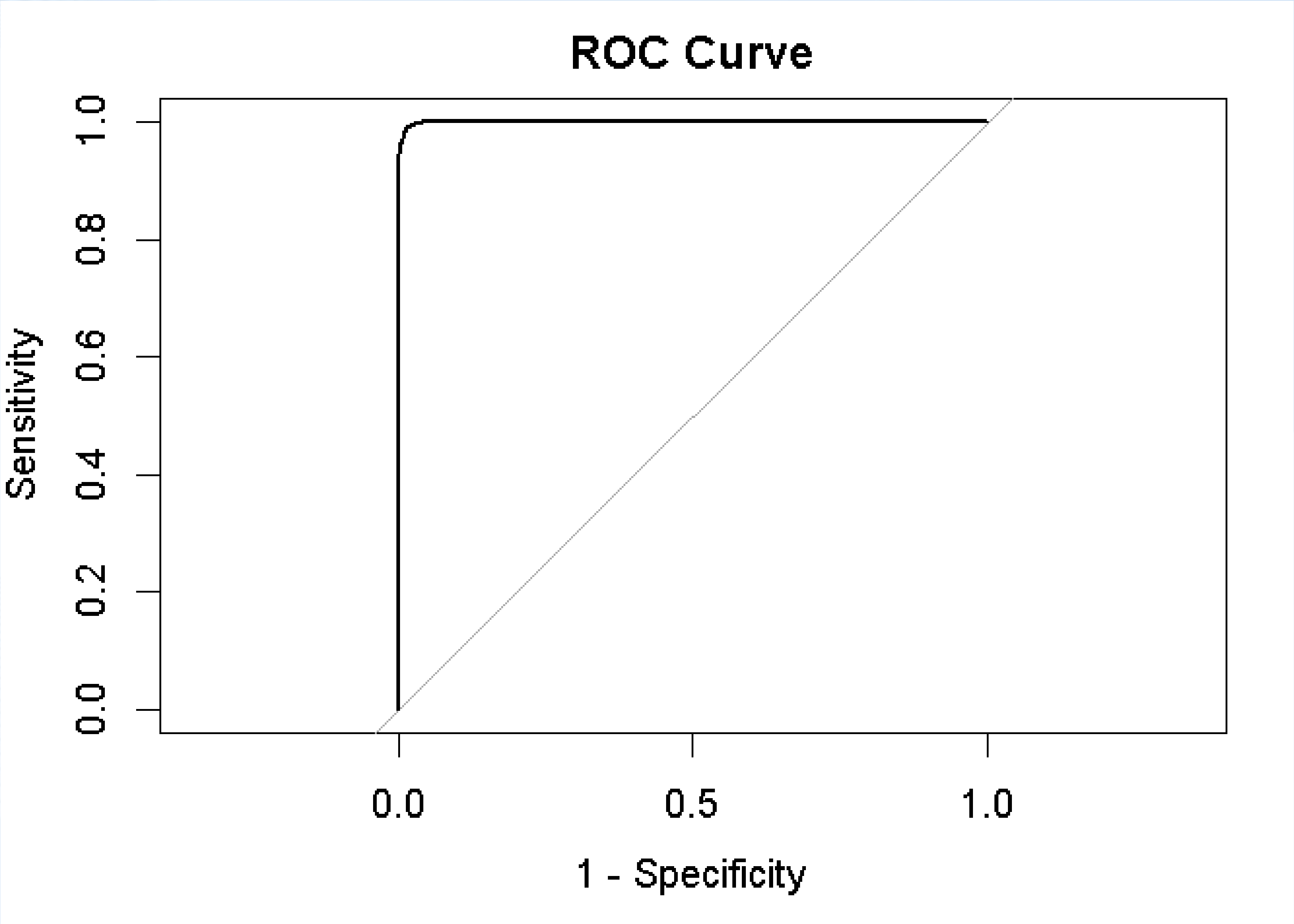
Therefore, we have sufficient evidence to say that model fits the data well at 5% significance level

Model adequacy checking

ROC curve method

- The ROC curve shows that the model is performing exceptionally well, with very high accuracy and discriminatory ability.
- The model achieved an AUC of 0.9994 which is extremely high, approaching 1.0, which indicates near-perfect discrimination.
- This means the model excellently distinguishes between the two classes of response (high vs. Low emissions) across all probability levels.

```
> plot(roc_obj, main = "ROC Curve")
> auc(roc_obj)
Area under the curve: 0.9994
>
```



Model adequacy checking

Hosmer and Lemeshow Goodness of fit test

- **Null hypothesis (H_0)**
The logistic regression model fits the data well.

- **Alternative hypothesis (**H_1)
The logistic regression model does **not** fit the data well.

- We got a p-value of 0.9998 is well above the 0.05 threshold indicating that the model fits the data very well.

```
         Hosmer and Lemeshow goodness of fit (GOF) test

data:  model$y, fitted(model)
X-squared = 0.041816, df = 4, p-value = 0.9998
```

Model adequacy checking

There is minimum multicollinearity because all standard GVIF values are less than 10.

As there is minimum multicollinearity and the model adequacy checking tests show that the model fits well, no interaction terms were considered.

```
> vif(logm)
                                 GVIF Df GVIF^(1/(2*Df))
FuelConsumptionCombL100km   81.395833  1        9.021964
as.factor(FuelType)         83.122732  2        3.019465
as.factor(VehicleClass)      1.884700  2        1.171684
as.factor(Transmission)      1.653974  3        1.087480
```

Final Adequate Model

$$\ln\left(\frac{\hat{P}}{1-\hat{P}}\right) = -175.434216.6041 \cdot \mathrm{FuelConsumptionCombL100km}$$
$$+ 21.3125 \cdot \mathrm{Fuel}_D - 70.7008 \cdot \mathrm{Fuel}_E$$
$$- 1.2783 \cdot \mathrm{VClass}_M - 1.5818 \cdot \mathrm{VClass}_S$$
$$+ 1.4475 \cdot \mathrm{Trans}_{AM} + 0.3211 \cdot \mathrm{Trans}_A - 0.5468 \cdot \mathrm{Trans}_V$$

$\hat{P}$ is the estimated proportion for high co2Emissionsgkm

Interpreting model parameters

- When fuelconsumptioncombl100km increases by one unit, the log odds of high co_2 emission increases by 16.6041,holding other variables constant

- Odds of having high co_2 emission for those who had used diesel as the fuel type is times 1.8 * 109,the odds of having high co_2 emission for those who had used gasoline ,holding other variables constant

- Odds of having high co_2 emission for those who had used gasoline as the fuel type is times 5.06 * 1030,the odds of having high co_2 emission for those who had used ethanol ,holding other variables constant

- Odds of a vehicle having high co_2 emission for large vehicle class is 3.5905 times the odds of a vehicle having high co_2 emission for medium vehicle class,holding other variables constant.

Interpreting model parameters

- Odds of a vehicle having high co_2 emission for large vehicle class is 4.8637 times the odds of a vehicle having high co_2 emission for small vehicle class,holding other variables constant.

- Odds of a vehicle having high co_2 emission for transmission AM is 4.2524 times the odds of a vehicle having high co_2 emission for transmission manual,holding other variables constant.

- Odds of a vehicle having high co_2 emission for transmission Automatic is 1.3786 times the odds of a vehicle having high co_2 emission for transmission manual,holding other variables constant.

- Odds of a vehicle having high co_2 emission for transmission manual is 1.7277 times the odds of a vehicle having high co_2 emission for transmissionVariable,holding other variables constant.

- The intercept estimate of -175.4342 represents the log odds of a vehicle having high co_2 emission when all categorical predictor variables are at their reference level (fuel type=Gasoline,vehicle class=Large,transmission type=Manual)and continuous predictor variables(Fuel Consumption Comb (L/100km)=0) are equal to zero.

Summary and Conclusion

Summary

•**Dataset:** Vehicle dataset containing EngineSizeL, FuelType, Transmission, VehicleClass and FuelConsumption.

•**Response variable:** CO_2 emissions (continuous and categorized: Low vs High).

•**Objective:** Identify key vehicle characteristics that influence CO_2 emissions.

•**Methods used:**

Multiple Linear Regression → Predict continuous CO_2 emissions.

Logistic Regression → Classify vehicles CO2 emission into Low vs High categories.

•**Focus:** Understand influence of fuel consumption, fuel type, cylinders, transmission, and vehicle class.

Summary

Multiple Regression Model	Logistic Regression Model
FuelconsumptionComb100km, Cylinders, FuelType, Transmission, VehicleClass	FuelconsumptionComb100km, FuelType, Transmission, VehicleClass
Error analysis / model adequacy: checked assumptions (normality, homoscedasticity, leverage).	Evaluated with Hosmer–Lemeshow test ($p = 0.9998 > 0.05$) → model fits data very well. ROC curve: AUC = 0.9994 → near-perfect discrimination between high vs. low emission categories.
Tried log transformation, centering variables, and polynomials → no improvement. Adding Fuel Consumption × Fuel Type interaction improved model fit.	No interaction terms considered since model was already adequate.
Interaction term caused high multicollinearity (high VIF). But it was kept due to better predictive performance	Final model considered adequate and robust for classification.

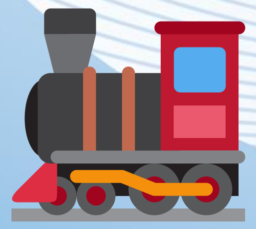
Conclusion

Fuel Consumption → strongest driver of CO_2 emissions.

Diesel fuel effect:

- *Multiple regression* → negative coefficient (lower emissions per unit fuel than gasoline).
- *Logistic regression* → positive coefficient (higher odds of high emissions, due to larger/heavier vehicles using diesel).

Transmission: Automatic/AMT → higher emissions compared to Manual.



Cylinders, vehicle class → influence CO_2 but mainly through their impact on fuel consumption.

Implications and Takeways

Optimizing **fuel consumption** is the most practical way to reduce CO_2 emissions.

Fuel type optimization (developing low-emission fuels) is more realistic than altering cylinders or vehicle classes.

Contradiction in diesel results shows the difference between:

- Modeling a continuous outcome (emission level).
- Modeling a categorical outcome (high vs. low emission).

Results align with real-world trends → diesel vehicles may emit less CO_2 per unit fuel, but their larger size/consumption makes them more likely to be high emitters.

Challenges and Limitations

Multicollinearity → Engine Size, Cylinders, Fuel Consumption highly correlated.

Class imbalance → Some categories underrepresented, possible bias.

Missing factors → Vehicle weight, aerodynamics, hybrid/electric systems not included → model oversimplified.

Model assumption issues → Deviations from residual normality, heteroscedasticity, high leverage points; limited knowledge to correct → potential inaccuracy despite high R^2.

Thank you